

Yihan (Yuki) Yang

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EDUCATION

University of Washington

Seattle, WA

First-year PhD student of the Paul G. Allen School of Computer Science & Engineering

Sep 2025 – Present

- Advisor: Michael Ernst

Harvey Mudd College (HMC)

Claremont, CA

B.S. in Joint Computer Science and Mathematics

Aug 2021 – May 2025

- Cumulative GPA: 3.97/4.0, Major GPA: 3.97/4.0, Dean's List 5 semesters
- Graduated with High Distinction, Honors in Mathematics, and Honors in Computer Science
- Related Coursework: Programming Languages, Database, Algorithm, Operations Research, Computability and Logic, Discrete Math, Abstract Algebra, Real Analysis, Algebraic Geometry, Representation Theory, Mathematics in Big Data, Intermediate Probability, Number Theory and Cryptography, Information Theory and Artificial Learning

PUBLICATIONS

- Mira Kaniyur, Ana Cavalcante-Studart, **Yihan Yang**, Sangeon Park, David Chen, Duy Lam, and Lucas Bang. 2024. *Interprocedural Path Complexity Analysis*. In Proceedings of the 33rd ACM SIGSOFT International Symposium on Software Testing and Analysis (**ISSTA 2024**). Association for Computing Machinery, New York, NY, USA, 162–173. <https://doi.org/10.1145/3650212.3652118>
- **Yihan Yang**, Mohit Tekriwal, John Sarracino, Matthew Sottile, Ignacio Laguna. 2025. *Towards Verified Linear Algebra Programs Through Equivalence* accepted at Coq for Programming Languages (**CoqPL**) in POPL2025
<https://popl25.sigplan.org/details/CoqPL-2025-papers/3/Towards-Verified-Linear-Algebra-Programs-Through-Equivalence>
- (poster) Mira Bhagirathi Kaniyur, Ana Cavalcante-Studart, **Yihan Yang**, Sangeon Park, David Chen, Duy Lam, and Lucas Bang. 2024. *Path Complexity Analysis for Interprocedural Code*. In Proceedings of the 2024 IEEE/ACM 46th International Conference on Software Engineering: Companion Proceedings (**ICSE-Companion '24**). <https://doi.org/10.1145/3639478.3643527>

PRESENTATIONS, and CONFERENCES

- Presented a 20 min talk at **CoqPL 2025** and attended **POPL 2025**, travel funded by SIGPLAN PAC
- Worked as a student volunteer and jointly delivered a 20 min talk at **ISSTA 2024**, travel funded by ACM-W and HMC.
- Presented “Interprocedural Path Complexity Analysis” at 2023 USC REU poster presentation, HMC summer CS research open house (July 2023), and HMC summer research poster presentation (Sep 2023).
- Presented “Verified Linear Algebra Programs Through Equivalence” at the LLNL summer poster session.
- Attended **OPLSS 2025**, travel funded by both OPLSS fellowship and UW.
- Attended the Joint Mathematics Meeting (**JMM 2024**) to gain conference experience and familiarize with cutting-edge mathematics.
- Attended PLMW workshop and **SPLASH 2024**, selected to receive travel grant.

AWARDS and EXAMS

- Wing and Ellen Tam Award, HMC (2025): given to student team for demonstrating excellence in software design and development.
- Courtney S. Coleman Prize, HMC (2023): outstanding performance in mathematics as a sophomore (3 out of Class of 2025)
- Scored 10/10 on the Actuary Probability (P) Exam (2023)
- Qualified for Math Prize for Girls (2020), top 300 girls nationwide in AMC12

RESEARCH EXPERIENCES

Lawrence Livermore National Lab (LLNL)

Livermore, CA

Undergrad Computing Scholars Intern

June – Aug 2024

- Advisors and PIs: John Sarracino, Mohit Tekriwal, and Ignacio Laguna.
- Utilized Rocq proof assistant and Mathcomp library to verify the correctness of classical and modified Gram-Schmidt in real numbers. Mechanized helper lemmas that are required for the proof of Gram-Schmidt using Mathcomp library; QEDed 30+ proofs in Rocq with the longest proof around 250 lines of code.
- Implemented Gram-Schmidt algorithms in Rocq to assist in proving equivalences and correctness.
- Co-wrote the paper “Towards Verified Linear Algebra Programs Through Program Equivalence”; presented at CoqPL 2025.

ALPAQA Lab Path Complexity Team, HMC

Claremont, CA

Undergrad Researcher, Advisor: Lucas Bang

May 2023 – Sep 2024

- Worked on quantifying the difficulty of automated testing using path complexity.
- Optimized existing code on path complexity in Metrinome (<https://github.com/hmc-alpaqa/metrinome>), and established theoretical improvements on the algorithm of path complexity calculation using concrete mathematics and algebraic combinatorics; made the algorithm 1000x faster.
- Analyzed different algorithms performance by implementing test cases and performing ablation studies; created a Python script to prevent caching conflicts between different test materials.
- Co-wrote paper and co-designed poster: “Interprocedural Path Complexity Analysis”; published and presented at ISSTA 2024.

Bee Lab, HMC

Claremont, CA

Undergrad Researcher, Advisor: Matina Donaldson

Jan – May 2023

- Modeled ant foraging behavior to analyze trade-offs between quality and distance
- Elaborated on the existing model and utilized Python and R for extensive result visualization; conducted parameter sweeps to analyze the influence of each variable on model performance and identified key parameter relationships to improve the model.
- Grouped and quantified parameters using interdisciplinary skills in economics, mathematics, and biology.

WORK EXPERIENCES

Webflow

San Francisco, CA

Contingent Worker at Webflow Labs

July – Sep 2025

- Worked with the Webflow R&D team and vendor Osmosis AI to validate LLM-generated outputs against Webflow Design Language.
- Built a Chrome Extension that utilizes Gemini API for automate training data generation.
- Built a visualizer to display LLM-generated outputs alongside rendered screenshots and ground-truth HTML, enabling clearer comparison and analysis of test results.

PROJECTS

Clinic Project (Webflow Team), HMC

Claremont, CA

Team Lead (Spring)/Team member (Fall)

Aug 2024 – May 2025

- Work with a team of five Harvey Mudd seniors to address the project sponsored by Webflow as part of the HMC Clinic program.
- Conducted user experience analysis by exploring Webflow's platform to identify key areas for improvement.
- Design and develop a Chrome extension that displays and manages the CSS of selected components of websites, using HTML/CSS, JavaScript, and React.
- Engage in regular communication with Webflow's liaison, faculty advisor, and team members, refining project management and collaborative problem-solving skills

Mathematical Contest in Modeling (MCM 2022)

Claremont, CA

Participant

Feb 2022

- Utilized Bitcoin price data to maximize profit by creating a price forecasting and strategic trading model.
- Applied ARIMA and LSTM to predict the price from past data and created three new models for trading strategies that have different dependencies on risk (code in Python); did sensitivity analysis on the different models.

TEACHING EXPERIENCES

Computer Science Department, HMC

Claremont, CA

Tutor and Grader

Aug 2023 – May 2025

- Tutored and graded "Computability and Logic" (Fall 2023 and Spring 2024) and "Algorithms" (Fall 2024 and Spring 2025); both are CS major requirements with ~80 students each.
- Tutored and guided students on homework by hosting tutor hours (four hours per week) and answering any course-related questions; each tutoring session has 10-15 attendees.
- Contributed to the grader and tutor meetings by sharing experiences and improving tutor strategies.
- Collaboratively graded homework, gave constructive feedback on proof writing, and handled regrade requests.

Math Department, HMC

Claremont, CA

Tutor and Grader

Aug 2022 – May 2025

- Graded Single and Multivariable Calculus (Fall 2022), Real Analysis (Spring 2023), Discrete Math (Fall 2023), Abstract Algebra (Spring 2024), and Number Theory (Spring 2025).
- Graded homework, including leaving suggestive comments on students' work, identifying logical flaws in proofs, and providing constructive feedback.
- Produced accurate solution keys (LaTeX) for Calculus class for other graders to reference.

SKILLS

Programming: Python, C++, R, Racket, Java, Haskell, Rocq, Prolog, SQL, JavaScript/HTML/CSS, React

Tools: GitHub, Mathematica, AMPL with Gurobi, Stata, Solidworks, ArcGIS, HMC machine shop certified

Language: English, Mandarin (proficient)

Fun: badminton, travel, photography